

ICT2217 Network Security
Lab Exercise 1.1: MAC Flooding Attack and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against MAC flooding attack

Required Hardware

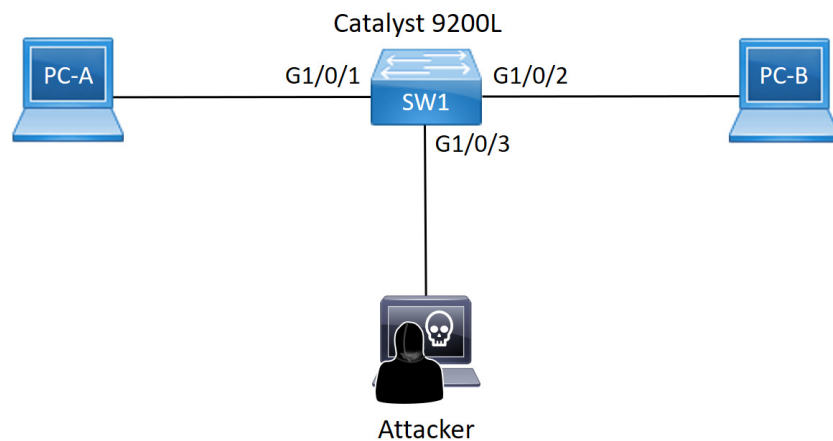
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-DB9 serial adapter + DB9-to-RJ45 console cable
 - USB-to-RJ45 console adapter
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

PART 1: Conducting MAC Flooding Attack using macof

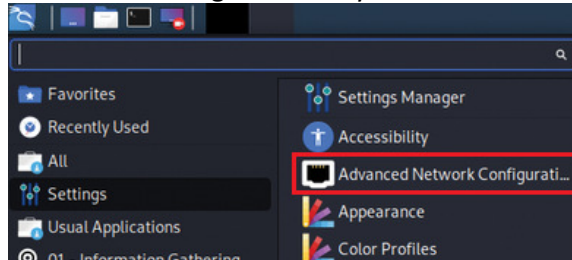
1.1 Set up the network topology with all switchports in default VLAN 1 as shown:



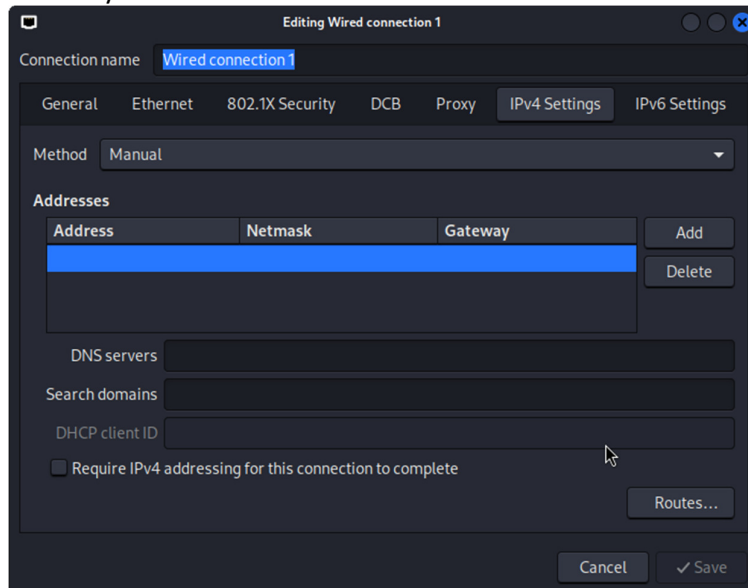
1.2 Boot up PC-A, PC-B and Attacker's PC in Kali Linux and configure suitable IP addresses manually.

Steps to configure IP address in Kali Linux:

- A. Click on the Kali icon at the top-left corner of the screen to bring up the menu and select Settings followed by Advanced Network Configuration as shown:



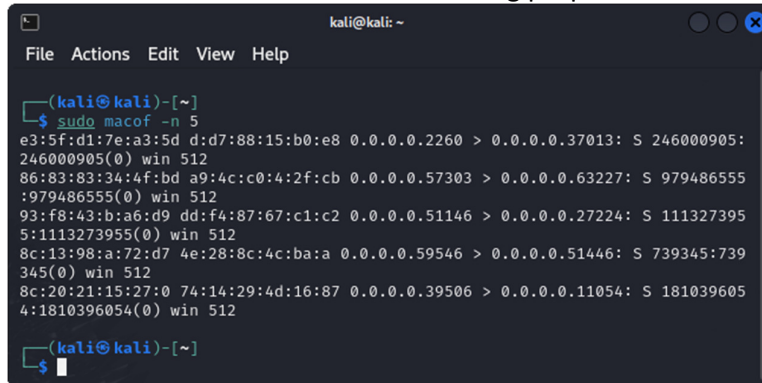
- B. On the Network Connections pop-up box, select the Wired connection to edit. Click on the IPv4 Settings tab, change Method from Automatic (DHCP) to Manual, and click the Add button to enter the IP address, Netmask and Gateway as shown:



- 1.3 Verify that all PCs can ping one another successfully to ensure that your setup is working correctly.
- 1.4 Before commencing MAC flooding attack, connect a laptop to the console port of the switch to observe the MAC address table.

```
SW1# sh mac address-table dynamic vlan 1
```

- 1.5 At the attacker's PC, start a terminal and launch macof to send 5 Ethernet frames with random source MAC addresses for testing purpose as shown:



```

kali@kali: ~
File Actions Edit View Help

(kali@kali)~$ sudo macof -n 5
e3:5f:d1:7e:a3:5d d:d7:88:15:b0:e8 0.0.0.0.2260 > 0.0.0.0.37013: S 246000905:
246000905(0) win 512
86:83:83:34:4f:bd a9:4c:c0:4:2f:cb 0.0.0.0.57303 > 0.0.0.0.63227: S 979486555:
979486555(0) win 512
93:f8:43:b:a6:d9 dd:f4:87:67:c1:c2 0.0.0.0.51146 > 0.0.0.0.27224: S 111327395:
5:111327395(0) win 512
8c:13:98:a:72:d7 4e:28:8c:4c:ba:a 0.0.0.0.59546 > 0.0.0.0.51446: S 739345:739
345(0) win 512
8c:20:21:15:27:0 74:14:29:4d:16:87 0.0.0.0.39506 > 0.0.0.0.11054: S 181039605:
4:181039605(0) win 512

(kali@kali)~$

```

WARNING: Disable Wi-Fi to prevent attacking other networks through Wi-Fi which may get you into trouble.

- 1.6 Show the MAC address table again. What difference do you observe as compared to Step 1.4?

- 1.7 Now, clear the MAC address table:

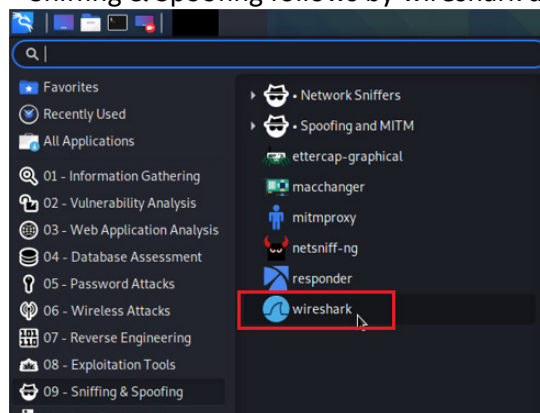
```
SW1# clear mac address-table dynamic
```

- 1.8 Next, modify the parameters of macof to conduct MAC flooding attack. Refer to <https://man.archlinux.org/man/macof.8.en> on the use of macof.

- 1.9 While the attack is in progress, note the remaining MAC address table space available:

```
SW1# sh mac address-table count
```

- 1.10 When done, start Wireshark at attacker's PC by clicking on the Kali icon and select 09 – Sniffing & Spoofing follows by wireshark as shown:



Note: If your macof is still running, stop it by entering Ctrl-C to prevent crashing Wireshark.

- 1.11 From PC-A, ping PC-B. Can your Wireshark capture the ping packets? Why or why not? Explain.
- 1.12 If you could not capture the ping packets in Step 1.11, troubleshoot the problem so that you are able to capture the ping packets successfully after conducting the MAC flooding attack.

PART 2: Defending against MAC Flooding Attack using Port Security

- 2.1 For your learning benefit, try Part 2 with different Port Security options and observe the effects.
- 2.2 When ready, show the MAC address table before MAC flooding attack.
- 2.3 Launch macof to conduct MAC flooding attack again.
- 2.4 What happen to the switchport connected to attacker's PC when launching MAC flooding attack?
- 2.5 Show the MAC address table again. What do you observe as compared to Step 2.2?
- 2.6 Do you think Port Security is a good defense against MAC flooding attack? Explain.